

LIVERMORE, CA, USA

WMO: 724927

Lat: 37.693N

Lon: 121.814W

Elev: 393

StdP: 14.49

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 23285

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.9	33.4	16.2	12.7	51.8	20.7	15.7	52.0	20.0	51.1	17.7	53.3	1.6	70	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.2	99.0	67.4	95.0	65.9	91.1	64.9	69.4	93.3	67.8	90.4	66.2	87.5	9.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	81.0	75.9	59.2	76.1	72.5	57.8	72.4	70.5	33.6	93.4	32.1	90.2	31.0	87.9	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.2	17.7	16.2	DB	26.9	105.5	2.0	2.9	25.5	107.6	24.3	109.3	23.2	110.9	21.7	113.0	
			WB	24.9	72.2	2.2	2.8	23.3	74.2	22.0	75.9	20.8	77.4	19.1	79.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.3	48.3	50.7	54.2	57.4	62.5	68.6	72.5	72.3	70.4	63.7	54.3	48.4	(6)
(7)		DBStd	10.41	4.88	4.87	5.22	5.71	5.88	6.41	5.85	5.02	5.75	5.35	5.48	5.08	(7)
(8)		HDD50	271	91	46	18	4	0	0	0	0	0	21	91		(8)
(9)		HDD65	2587	517	400	334	239	122	28	5	3	14	88	322	515	(9)
(10)		CDD50	4044	39	66	149	226	387	558	698	692	612	426	150	41	(10)
(11)		CDD65	884	0	0	1	10	44	136	237	230	176	49	1	0	(11)
(12)		CDH74	10910	0	2	57	269	742	1827	2703	2451	2058	761	40	0	(12)
(13)		CDH80	5288	0	0	3	79	316	922	1412	1236	1052	266	2	0	(13)
(14)	Wind	WSAvg	6.2	4.6	5.0	5.5	6.6	8.1	8.5	8.6	7.9	6.3	5.0	4.1	4.6	(14)
(15)	Precipitation	PrecAvg	14.9	2.9	2.6	2.2	1.1	0.4	0.1	0.0	0.1	0.2	0.8	2.0	2.4	(15)
(16)		PrecMax	32.4	6.7	7.3	6.7	4.6	2.0	0.7	0.7	0.9	1.5	3.6	5.4	7.0	(16)
(17)		PrecMin	6.4	0.3	0.1	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	(17)
(18)		PrecStd	5.2	2.0	1.9	1.7	1.0	0.5	0.2	0.1	0.2	0.4	0.9	1.6	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.8	72.8	79.5	87.9	95.0	102.0	103.7	101.9	100.6	92.7	78.6	67.2	(19)
(20)		0.4%	MCWB	51.9	56.4	60.5	62.8	65.9	68.8	69.3	68.1	66.5	63.7	58.5	54.9	(20)
(21)		2%	DB	64.2	68.5	74.9	81.5	88.4	95.8	97.8	96.7	96.2	86.5	73.9	63.4	(21)
(22)		2%	MCWB	52.2	55.2	58.7	61.5	63.5	66.6	67.6	66.4	65.2	61.1	57.5	53.0	(22)
(23)		5%	DB	60.9	64.4	70.5	75.8	82.4	90.7	93.0	91.8	91.4	82.2	69.0	60.8	(23)
(24)		5%	MCWB	52.2	54.0	56.7	58.7	60.9	64.9	66.4	65.6	63.8	59.3	55.4	52.8	(24)
(25)		10%	DB	57.6	61.0	66.2	70.5	76.4	84.7	88.2	87.0	86.4	77.8	65.2	58.3	(25)
(26)		10%	MCWB	50.9	52.2	54.7	56.4	59.1	62.9	65.3	64.6	62.5	58.1	53.8	51.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.9	59.7	63.0	65.5	68.2	70.9	72.7	71.2	70.9	66.0	62.0	59.6	(27)
(28)		0.4%	MCDWB	62.0	67.2	76.3	81.9	91.1	97.0	99.2	96.7	89.6	85.0	72.9	63.4	(28)
(29)		2%	WB	55.4	57.4	60.1	62.4	64.9	68.2	69.3	68.4	67.3	63.5	59.7	57.2	(29)
(30)		2%	MCDWB	59.8	65.0	71.7	78.7	85.2	92.5	92.9	91.7	89.7	81.1	69.0	60.8	(30)
(31)		5%	WB	53.6	55.3	58.0	59.6	62.2	66.0	67.4	66.6	65.5	61.7	57.9	55.1	(31)
(32)		5%	MCDWB	57.5	62.5	68.2	73.4	79.7	87.9	89.8	87.6	87.0	77.3	65.4	58.5	(32)
(33)		10%	WB	51.8	53.5	55.8	57.3	59.8	63.9	65.8	65.2	63.9	59.9	56.2	52.8	(33)
(34)		10%	MCDWB	56.5	59.7	64.2	68.8	74.1	82.8	86.6	84.8	83.4	73.3	63.0	56.2	(34)
(35)	Mean Daily Temperature Range	MDBR	18.1	19.2	21.5	22.6	23.9	27.4	29.2	28.9	28.5	25.9	21.4	18.3	(35)	
(36)		5% DB	MCDDBR	23.0	25.5	29.4	32.9	35.1	37.1	36.1	36.7	37.6	34.2	27.1	22.0	(36)
(37)			MCWBR	13.8	14.9	15.9	15.8	15.0	14.5	13.2	13.4	14.6	14.8	13.9	13.3	(37)
(38)			MCDDBR	16.7	21.1	26.5	29.3	32.1	34.3	33.6	32.9	32.6	28.6	22.0	14.3	(38)
(39)		5% WB	MCWBR	11.3	13.2	15.3	15.0	14.6	14.1	13.2	12.6	13.2	13.2	12.6	9.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.325	0.339	0.359	0.365	0.360	0.365	0.350	0.350	0.333	0.325	0.324	(40)	
(41)		taud	2.527	2.518	2.459	2.383	2.388	2.410	2.398	2.431	2.426	2.507	2.510	2.520	(41)	
(42)		Ebn at Noon	275	289	293	292	290	291	288	291	285	281	270	265	(42)	
(43)		Edh at Noon	25	28	33	37	38	37	37	35	33	28	25	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	749	1059	1503	1959	2312	2558	2480	2224	1843	1338	901	666	(44)	
(45)		RadStd	109	128	140	164	138	99	69	71	63	81	56	98	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.99	N/A	N/A	N/A	N/A	N/A	-50	-234	+356	+108	(46)
(47)	Regional (44 neighbors)	N/A	N/A	-2.11	N/A	N/A	N/A	N/A	-198	+243	N/A	(47)

Nomenclature: See separate page